

Chapter 8

Cultural and Integrated Control of *Botrytis* spp.

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Abstract Epidemics of *Botrytis*-incited grey mould are common in open fields, orchards and greenhouses. These infections are promoted by high humidity and the presence of a film of water on susceptible plant organs and those conditions may be manipulated to prevent infection. Traditionally, heating greenhouses was a popular means of controlling the humidity in those structures and this practice remains popular in some temperate regions. However, the cost of active heating has forced farmers in some regions to abandon this disease-management strategy, which has led to the increased incidence and severity of grey mould, as susceptible organs of the crop plants remain wet for longer periods of time. Cultural methods for controlling *Botrytis*-incited disease include reducing the planting density, managing the crop canopy to allow for the aeration of the crop or susceptible organs via passive and active ventilation, fertigation with increased levels of potassium and calcium and reduced amounts of nitrogen, the use of soil mulch and passive solar heating of unheated greenhouses, avoiding harvesting on rainy days, and timing fungicide applications for the optimal protection of fresh harvest wounds and so that the crop remains wet for shorter periods of time. Row and field positioning and direction can affect grey mould as these factors affect local air movement and temporal temperature changes. A combination of treatments can provide better grey mould suppression than individual treatments and appropriate integration of control measures can provide sufficient disease control with minimal use of chemical fungicides.

Keywords Agrotechnical disease control • Integrated disease management • Plant nutrition • Resistance

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8.1 Introduction

Although the limiting factors for the germination of *Botrytis* spp. conidia and grey mould infection are generally the presence of liquid water and high humidity, the management of *Botrytis*-induced diseases can vary in different agro-systems. The initial *Botrytis* spp. inoculum in various crops may originate from inside or outside the field, as shown for *B. cinerea* in greenhouses (Korolev et al. 2006). In some crops, grey mould epidemics have been described as polycyclic (Ben Kalifa et al. 2012; Decognet et al. 2009; Elmer and Michailides 2004). Additionally, the common targets for efforts aimed at suppressing *Botrytis* spp. under field conditions in various crops are the possibility of limiting the presence of water, growing cultivars/crops with greater resistance to the pathogen and manipulating the availability of nutrients to the plants to decrease host susceptibility. However, agro-technical means for suppression of epidemics may vary according to limitations imposed by the local growing conditions and what host plants can tolerate without any significant yield loss.

Practices that allow air to move within the crop canopy, reduce the relative humidity around the plant, limit the presence of a film of water and increase resistance will be described below. General strategies for reducing the general humidity within a crop include planting on relatively high ground, allowing the wind to move along the rows and mulching with polyethylene or straw. Heating and ventilating greenhouses can be a powerful technique for controlling humidity inside those structures and suppressing disease suppression. But, this may not be economically viable when energy prices are high. Nutrition can affect the susceptibility of plants to *Botrytis* spp. and those effects will also be discussed in this chapter. In our discussion below, the plant hosts of *Botrytis* spp. are divided into two groups: open-field (and orchard) crops and greenhouse-grown crops. In this chapter, we describe cultural means of *Botrytis* control as stand-alone practices in various crops, as well as combined means of control in some specific pathosystems.

8.2 Host Characteristics

Botrytis cinerea is a major pathogen in orchards and vineyards. Various means of combating fruit infections have been explored and implemented; these methods have generally relied upon host characteristics and canopy management (Elmer and Michailides 2004). The morphological, anatomical and chemical characteristics of grape cultivars were measured in an attempt to correlate grey mould susceptibility with host characteristics. In that work, the number of naturally occurring pores in a leaf was found to be negatively correlated with resistance to *B. cinerea* (Eibach 1994; Mlikota Gabler et al. 2003). Cultivar resistance was also associated with higher cuticle and wax contents, but not induced or constitutive antifungal host defence mechanisms (Mlikota Gabler et al. 2003). In some raspberry cultivars, the stigmatic fluid inhibited *B. cinerea*. This mechanism allows those plants to avoid

latent infections, which might develop into acute infections (Williamson and Jennings 1992). This observation suggests that cultivar selection can play a role in *Botrytis* management strategies.

The effects of root depth and water uptake on skin splitting and grey mould have been studied on wine grapes in France (Ribèreau-Gayon et al. 1980). Berries on deep-rooted vines were found to be much less susceptible to splitting and grey mould than those on shallow-rooted vines. Rootstocks also had a significant effect on the extent of fruit micro-cracking in sweet cherries and this rootstock effect is believed to be due to differences in their uptake of soil moisture (Cline et al. 1995). Thus, not only genetic resistance of cultivars, but also the nature of the rootstock can affect the outcome of the interaction between *Botrytis* and its host.

8.3 Canopy Management and Sanitation

Dense grape canopies are associated with increased wetness and more severe grey mould epidemics. Improving air circulation in the bunch zone, reducing relative humidity in the canopy and exposing more fruit bunches to light (Savage and Sall 1982) reduced bunch rot (Smithyman et al. 1997). Leaf removal affects the micro-climate around the bunches of grapes. The removal of leaves from the fruiting zone of vines significantly reduced the incidence of *Botrytis* epidemics in grapes (English et al. 1989, 1993; Gubler et al. 1987; Percival et al. 1994; Thomas et al. 1988; Zoecklein et al. 1992). Following leaf removal, exposed Riesling grapes developed large amounts of epicuticular wax and cuticle, which resulted in significantly lower levels of grey mould (Percival et al. 1993).

Vines trained along horizontal bilateral cordons had improved exposure to light and a lower incidence of *Botrytis* than vines in a 'high' cordon system supported by a one-wire trellis (Redl 1988). A higher incidence of grey mould was observed in the Pergola system, whereas vines pruned and trained to the Guyot system had the lowest levels of disease (Cargnello et al. 1991). Infection of Cabernet Sauvignon clusters by *B. cinerea* after veraison was significantly influenced by cluster compactness (Fermaud et al. 2001a, b; Vail and Marois 1991). Reduction of berry-to-berry contact within the bunch reduced the development of *B. cinerea* (Martin 1990).

Botrytis spp. also infect various legume crops (Davidson et al. 2004). Grey mould of lentils is caused by both *B. cinerea* and *B. fabae* (Bailey et al. 2000). Fava beans are infected by both *B. fabae* and *B. cinerea*, but *B. fabae* is more important in causing chocolate spot in this crop (Sundheim 1973; Mansfield 1980). There are some effective cultural methods for controlling humidity-promoted diseases in annual open-field legume crops (Davidson et al. 2004). A dense canopy encourages the development of lentil grey mould, but there are several agronomic practices that can be used to avoid a dense canopy (Bretag and Mebalds 1987), including the adjustment of sowing dates and sowing density, increasing the amount of space between rows and employing appropriate weed control (Bayaa and Erskine 1998). Dense plantings of fava bean create a high-humidity environment, which should

also be avoided (Jellis et al. 1998). An association between early sowing and chocolate spot was noted in Syria (Hanounik and Robertson 1988). In common bean, cultural practices can provide disease reduction i.e. reduced planting densities, careful use of fertilisers to prevent the development of a dense crop and judicious watering all help to reduce the onset of disease (Jarvis 1991).

The removal of necrotic tissue before onset of epidemics led to significant reductions in the number of *Botrytis* spp. conidia in the air and incidence *Botrytis* infections (Köhl et al. 1992). Similarly, the removal of kiwifruit flowers from male vines (an important source of *B. cinerea* inoculum) led to reduced levels of stem-end rot in neighboring vines (Michailides and Elmer 2000). Removal of senescent floral tissues and aborted berries was found to reduce *B. cinerea* by approx. 30 % in Merlot grapes (Jermine et al. 1986). Similarly, pruning infection foci in tomato leaves stopped epidemics in greenhouses (Shtienberg et al. 1998). These results reveal an effective means of epidemic management that is, unfortunately, quite labour intensive.

8.4 Managing Environmental Conditions Inside the Greenhouse

8.4.1 Heated Greenhouses

Botrytis cinerea infects many vegetable, ornamental and other horticultural crops (Elad et al. 2004). The temperature range for grey mould development in those various crops is from 12 to 30 °C (Jarvis 1992), with an optimal range of 15–20 °C, although the pathogen is active at temperatures as low as 0 °C (Droby and Lichter 2004). Greenhouses may or may not be actively heated. The use of supplemental heating has been shown to result in lower levels of grey mould in greenhouse-grown vegetable and ornamental crops (Dik and Wubben 2004; Elad and Shtienberg 1995; Hausbeck et al. 1996; Jarvis 1992).

Studies on grey mould have suggested the use of heating or aeration and ventilation of the greenhouse (Dik and Wubben 2004; Eden et al. 1996; Elad and Shtienberg 1995; Jarvis 1992; Morgan 1984) to reduce humidity levels and the amount of dew on susceptible plant tissues. Heating greenhouses not only results in higher temperatures, but can also lead to higher vapour pressure deficits (lowered relative humidity; Dik and Wubben 2004). In heated geranium (*Pelargonium*) crops, it was demonstrated that periods of wetness lasting more than 4 h should be avoided (Sirjusingh and Sutton 1996) and that forced heated air can reduce the incidence of stem blight and sporulation (Hausbeck et al. 1996). In heated tomato greenhouses with three climate regimes, more grey mould was found in crops in greenhouses kept at 16 °C with no ventilation. Less tomato grey mould was observed in crops kept at 13 °C with ventilation and even less in those kept at 16 °C with ventilation (Morgan 1984). These results demonstrate the importance of ventilation for reducing humidity and preventing the conditions that promote infection.

8.4.2 Light Quality Management in Greenhouses

Crops susceptible to grey mould may be grown in polyethylene tunnels. Different types of polyethylene have different effects on the sporulation of *B. cinerea* (Reuveni et al. 1989). Light, especially near-UV (nUV) light, increases the sporulation of *B. cinerea*. In *Primula* and strawberry crops grown under polyethylene tunnels, the incidence of infection over two seasons was reduced when nUV blocking film (up to 405 nm) was used instead of a standard film (West et al. 2000). Several weeks after the inoculation of tomato plants with *B. cinerea*, conidia production was low under polyethylene films that absorbed nUV light (Nicot et al. 1996).

Spectrally modified polyethylene films were found to have a similar effect on the development of *B. cinerea* on greenhouse-grown tomato plants (Reuveni and Raviv 1992) and UV-absorbing polyvinyl chloride film had a similar effect in cucumber (Honda et al. 1977). Elad (1997) demonstrated an effect of far-red light on the production of conidia by field isolates of *B. cinerea*. A green pigmented polyethylene film reduced conidia production and infection in commercial greenhouses. However, Elad (1997) identified some field isolates of *B. cinerea* that were capable of producing conidia in the dark and were not affected by light quality. Since similar *B. cinerea* fungi exist under field conditions, it was concluded that the disease-suppressive effect of light-filtering polyethylene covers was definitely due to a change in the physiological status of the plants, that is, induced resistance in plants grown under conditions of change in light quality (Elad 1997).

8.4.3 Crop Density in Greenhouses

Increasing the air circulation in the field may reduce grey mould by reducing the amount of time that a water film is present on susceptible plant parts (Elad and Shtienberg 1995). Cultural methods such as reduced planting density and reduced canopy density can help to improve the environmental conditions in the field and limit the development of grey mould (Legard et al. 2000; Trolinger and Strider 1984). Similarly, reducing leaf wetness by moving from open-field production to unheated greenhouses dramatically reduced the incidence of strawberry grey mould (Xiao et al. 2001).

Crop density affects the movement of air in the canopy and grey mould severity. More cucumber grey mould was found on fruits in a cucumber crop with two stems per plant than in a crop with one stem per plant (Elad and Shtienberg 1995). Higher levels of grey mould were observed on floral parts of *Exacum affine* grown under crowded conditions as compared to less crowded conditions (Trolinger and Strider 1984) and similar results were observed in densely planted strawberry (Legard et al. 2000).

8.4.4 Stem Infection and Its Management

Grey mould develops at relative humidity (RH) levels of >90 % and *Botrytis* conidia germinate in liquid water (Jarvis 1992). However, grey mould can develop on tomato and sweet basil stem wounds under lower RH conditions. In these crops, optimal disease development has been documented at 75–85 % RH (Sharabani et al. 1999; O'Neill et al. 1997). *B. cinerea* infection at lower RH is possible due to moisture on the wound surface. This moisture originates from within the plant and promotes infection. The infection further develops inside the host tissue, where it is wet (Eden et al. 1996).

Lisianthus (*Eustoma grandiflorum*) is a cut flower crop that is susceptible to infection by *B. cinerea* (Shpialter et al. 2009), which causes grey mould on the harvested flowers and stem bases and on the stem stubs after harvest (Shpialter et al. 2009; Wegulo and Vilchez 2007), especially in unheated greenhouses. The lower leaves of lisianthus plants are often in close contact with the wet soil. When a barrier, in the form of a polyethylene sheet, was placed between the lower leaves and the wet soil, the severity of disease symptoms on those leaves was significantly reduced (Shpialter et al. 2009).

Infection of the stem base is likely to be affected by soil moisture, due to the proximity of the stem base to the soil. The location of irrigation drippers affects the moisture level of the soil around the stem base and is also likely to affect the infection of plants by *B. cinerea*. In lisianthus, when drippers were placed 20 cm below the soil surface instead of at the soil surface, the incidence of stem infection was drastically reduced. The sub-surface irrigation plots were characterized by lower RH and shorter dew periods than those observed in plots in which an aboveground drip irrigation system was used (Shpialter et al. 2009).

In a study of early blight (*Alternaria solani*) and Septoria leaf spot (*Septoria lycopersici*) in fresh market tomato crops, Mills et al. (2002) concluded that an observed reduction in foliar disease in mulched plots was associated with reduced splash dispersal and, in one of two trials, reduced leaf wetness. Strawberry plants grown in beds covered with black polyethylene had lower incidences of *B. cinerea*-incited rot than those grown in beds covered with either clear polyethylene or paddy straw (Saks et al. 1996). Reducing ambient humidity by covering the soil with polyethylene has also been suggested as a means for restricting grey mould development in greenhouse crops (Elad 2000; Elad and Shtienberg 1995).

In greenhouse-grown lisianthus, polyethylene mulch reduced RH and dew duration and drastically reduced the incidence of disease (Shpialter et al. 2009). The mechanisms by which lisianthus stem base infections are reduced through the use of polyethylene soil cover may be complex. The reductions in the RH and the amount of dew in the canopy contribute to the inhibition of the germination of *B. cinerea* conidia and limit the fungus's penetration of the host plant (Holz et al. 2004). Also, as shown in laboratory experiments, a physical barrier between the lower leaves and the wet soil affects conditions for the development of infection on the leaves near the stem base.

Grey mould of lisianthus stems can also serve as an example of a situation in which cultural methods and their combination can affect disease development. In-bed aeration in greenhouse-grown lisianthus, based on the use of perforated polyethylene sleeves and blowing unheated air, reduced disease incidence by changing the microclimate around the stem base (Elad et al. 2009b). As mentioned above, soil cover suppressed disease. Moreover, small planting holes in the polyethylene were superior to larger holes. This is probably due to the fact that when large holes are used, the lower leaves may come into contact with the wet soil and this moisture may promote *B. cinerea* infection of those leaves and the spread of that rot towards the stem. In contrast, when the planting holes are smaller, the lower leaves are less likely to come into contact with the wet soil. The disease control provided by whole-plant applications of chemical fungicides together with soil cover is not superior to that provided by soil cover alone. A polyethylene soil cover with small planting holes was as effective as chemical control. The combination of a polyethylene soil cover with small holes and chemical sprays directed at the plant stubs provided better disease suppression than either of the treatments alone (Elad et al. 2009b).

8.4.4.1 Effects of Polyethylene Soil Cover and Deep Irrigation on Foliar *B. cinerea* Infection

The use of polyethylene soil cover can suppress grey mould (Elad 2000), as mentioned above. For instance, the use of plastic sheets that have a white upper surface and a black lower surface increased yield and fruit size in strawberry and reduced the incidence of *B. cinerea* infection in that crop, as compared with non-mulched soil (Laugale et al. 2012). Polyethylene soil cover is associated with increased yields, thanks to the accumulation of heat in the root zone and increased soil temperatures (Flores Velasquez and Ibarra 1998).

Apart from the physical barrier between the soil and the aerial plant parts and the prevention of evaporation from the soil, Cohen et al. (2006) suggested a detailed explanation for the effect of polyethylene mulch on foliar late blight (*Phytophthora infestans*) of tomato and *Pseudoperonospora cubensis* in cucumber. They found that, in the early winter, covering the soil surface with polyethylene dramatically reduced the rate of infection, as compared with that observed in greenhouses in which the crops were grown in bare soil. Microclimate measurements showed that night-time soil and air temperatures were higher in the mulched greenhouses. On days when infection occurred in the control plots, air temperature reached the dew-point temperature in the control greenhouses, but not in the mulched greenhouses. Night-time soil temperatures and air temperatures near the ground were higher under the mulch and the difference between the air temperature and the dew point in the greenhouse was higher above the mulch, indicating a lower incidence of condensation on the leaves (Shtienberg et al. 2010; Cohen et al. 2006).

This temperature–dew effect explains part of the effect of polyethylene mulch on grey mould, but not the effect of deep irrigation, which also affected the duration of wetness of the lower leaves and the RH. Interestingly, plant tissues sampled in the

mulched plots had relatively high levels of N, P, K, Ca, and Mg, as compared to tissue samples collected from the unmulched plots. These nutritional differences might contribute to disease control by affecting the susceptibility of lisianthus tissues to *B. cinerea* infection (Shpialter et al. 2009).

8.4.4.2 The Sweet Basil–Grey Mould Test Case

Sweet basil (*Ocimum basilicum*) is an annual herb crop grown in greenhouses. Branches are harvested several times a season and this harvesting creates wounds that are susceptible to infection by *B. cinerea*. Stem infection can develop until the entire plant dies and a large amount of conidia are discharged into the air (Sharabani et al. 1999). Rot can also develop on the crop postharvest (Aharoni et al. 2010). Sweet basil is grown year-round in unheated polyethylene greenhouses, but the majority of grey mould damage is observed during the winter which is the rainy season in the region. In the absence of active heating, the conditions in sweet basil greenhouses are favorable for *B. cinerea*. Indeed, it was found that the limiting factor for grey mould development was the need for a high level of humidity in the greenhouse rather than the presence of inoculum. Regardless of crop age, the severity of grey mould observed on harvesting wounds was found to be related to seasonal microclimatic conditions.

Sharabani et al. (1999) found that rain outside the sweet basil greenhouses, which is naturally associated with increased RH in the greenhouse, was the dominant factor influencing the timing of grey mould epidemics. The cut ends of stems were found to be highly susceptible to infection soon after harvest; susceptibility diminished gradually and stem cuts inoculated 2 days after harvest were rarely diseased. In field experiments (Sharabani et al. 1999), disease severity did not change much when harvesting was done on dry days. However, disease outbreaks were found to coincide with harvests carried out on rainy days. It was found that if harvest was carried out during or immediately after a rain event, then the application of one fungicidal spray soon after harvest is needed and will provide adequate disease suppression. Avoiding harvesting during rain events is another good way to prevent stem grey mould (Sharabani et al. 1999). Applying the fungicide treatment immediately after harvesting (wounding) may help to reduce the amount of chemical fungicide needed, as well as lower the risk of contaminating the marketable crop with pesticide residues.

Plant spacing and foliage management can affect the RH inside the crop canopy and denser foliage is associated with higher levels of grey mould incidence (Legard et al. 2000; Trolinger and Strider 1984). Observations in commercial sweet basil plots have revealed that higher planting densities and restricted aeration contribute to epidemics (Elad et al. 2014b). In experiments, a planting density that was half that of the common practice was associated with lower disease levels, with no significant negative yield effect. This could be attributed to better aeration within the canopy at the initial stages of growth and to reduced amounts of receptive host tis-

sue. At a later stage, the branches harvested from the lower-density plots were less susceptible to *B. cinerea* infection, indicating that a higher level of plant tissue resistance was induced by the lower-density field conditions (Elad et al. 2014b). A combination of increased plant spacing and the use of a polyethylene soil cover synergistically improved the yield of sweet basil branches. Similarly, increasing the distance between plants in strawberry plots reduced the incidence of grey mould in those plots. Nevertheless, the fruit yield was negatively affected by this practice (Laugale et al. 2012; Daugaard 2003; Legard et al. 2000).

Aeration is an accepted practice for reducing the RH inside greenhouses. For instance, nocturnal ventilation can significantly reduce the level of humidity in a greenhouse and suppress tomato *B. cinerea* grey mould (Baptista et al. 2012). Aeration of sweet basil tunnels decreased the incidence of grey mould in those tunnels. Moreover, the combination of passive aeration and wider plant spacing improved the yield of sweet basil, as did the combination of wider plant spacing and polyethylene soil cover (Elad et al. 2014b).

8.4.5 Passive Heating of Polyethylene Greenhouses

We have developed a method to suppress foliar diseases by increasing the temperature inside greenhouses during day. Elevated day-time temperatures are obtained by closing the side walls of the greenhouses, which allows the temperature inside the greenhouse to rise above 25 °C in winter. The initial information on the effect of day-time elevated temperatures was published in the context of the tomato–*O. neolycopersici* pathosystem (Elad et al. 2008; Jacob et al. 2007) and the sweet pepper–*O. sicularis* pathosystem (Elad et al. 2009a).

Temperatures in polyethylene-covered sweet basil greenhouses that were kept closed for 6 h each day reached 42 °C on winter days. The incidence of grey mould in these greenhouses was significantly reduced and basil yield increased as a result of the passive day-time heating. A significant negative correlation was observed between the duration of temperatures above 30 °C and the incidence of grey mould. The suppressive effects of high day-time temperatures on the three diseases in the three plant species can be attributed to a direct effect of temperature on the pathogen inoculum, as well as an indirect effect involving the induction of resistance in the plants. New results from work with sweet basil (Elad et al. 2014b) indicate that the heat treatment has an indirect effect involving the induction of resistance in the host plants. In this case, plants exposed to a day-time heat treatment and later infected by *B. cinerea* or *S. sclerotiorum* exhibited less severe infections than inoculated plants that had not received the heat treatment (Elad et al. 2014a). Thus, it may be concluded that high day-time temperatures may reduce foliar disease. However, one needs to carefully verify that this heating has no deleterious effect on fruiting or negative interaction with other biotic stresses, such as humidity-promoted pathogens.

8.5 Plant Nutrition

Mineral nutrients are essential for plant growth, development and reproduction. Nutrients can also affect plants' susceptibility to pathogens (Engelhard 1989). Calcium is important for fruit and vine production (Ferguson 1984), affects various cell functions (Conway 1982) and increases resistance to disease (Volpin and Elad 1991; Conway et al. 1991). Calcium in plant tissues has been shown to reduce the severity of grey mould (Yermiyahu et al. 2006; Bar-Tal et al. 2001; Chardonnet and Doneche 1995; Volpin and Elad 1991).

Calcium reduces the leakage of exudates to the host surface, reducing the availability of those exudates to the pathogen (Volpin and Elad 1991). It can also directly inhibit the production of polygalacturonase by *B. cinerea* (Biggs et al. 1997). Ca^{2+} deficiency increases susceptibility to *B. cinerea* (Schwab et al. 1993). Incubating *B. cinerea* conidia in increasing concentrations of CaCl_2 decreased conidial germination and germ tube length (Chardonnet et al. 2000). Possible mechanisms by which high levels of calcium in plant tissues may reduce disease include the reduced emission of ethylene from the plant tissue, decreased susceptibility of plant membranes, particularly pectin constituents, to pathogens' hydrolytic enzymes and reduced leakage of nutrients to plant surfaces (Elad and Evensen 1995).

In strawberry, increasing fruit calcium content reduced the incidence of *Botrytis* infection (Cheour et al. 1990; Karp and Starast 2002; Wojcik and Lewandowski 2003). Nevertheless, when conventional fungicides were replaced with CaCl_2 to protect strawberries from *B. cinerea*, the CaCl_2 treatments did not reduce the epidemic (Erincik et al. 1998). When Ca^{2+} was applied to table grapes in the vineyard, resistance to *B. cinerea* increased and this resistance was correlated with increased levels of cellulose and both oxalate- and alkali-soluble pectins (Miceli et al. 1999).

High nitrogen rates enhance plant growth and foliage density. In general, increasing the nitrogen level is believed to result in increased susceptibility to *B. cinerea*. However, the results in this area are sometimes contradictory. Nitrogen supplied at 1.5, 3.8 and 6.0 g/m² resulted in a quadratic increase in grey mould incidence on chrysanthemum flowers (Hobbs and Waters 1964). In contrast, Verhoeff (1968) found that the susceptibility of soil-grown tomatoes to grey mould increased with decreasing levels of nitrogen in the soil. Hoffland et al. (1999) found a linear correlation between the leaf C:N ratio and the susceptibility of tomato to *B. cinerea*. They attributed that correlation to levels of available soluble carbohydrates.

Increasing calcium concentrations in fertigation fluid reduced the intensity of conidiation of *B. cinerea* on sweet basil stems and the incidence of infected plants (Yermiyahu et al. 2006). Furthermore, increased nitrogen increased the susceptibility of sweet basil to *B. cinerea*; this effect was related to the indirect negative influence of N on Ca in the canopy (Yermiyahu et al. 2006). Similarly, increased N levels have been shown to increase the severity of grey mould in chrysanthemum (Hobbs and Waters 1964), but decrease the incidence of the disease in tomato (Verhoeff 1968). High levels of nitrogen result in increased plant growth and the increased incidence *B. cinerea* on fruits in storage (Pertot and Perin 1999; Prasad and Spiers 1991). In sweet basil, higher nitrogen levels in the fertigation solution

resulted not only in more grey mould infection, but also in increased severity of white mould (*Sclerotinia sclerotiorum*; Elad and co-workers, unpublished). These findings indicate that limiting nitrogen fertilization can minimize grey mould, but that would also be expected to affect yields.

Similar nutrition guidelines are commonly suggested for open-field crops such as legumes. Avoiding a dense canopy in lentil by avoiding high N levels decreases grey mould (Bayaa and Erskine 1998). A higher incidence of chocolate spot was observed in plants fava bean grown with increased applications of N fertiliser (Hegab and Beshir 1994). However, in fava bean, the use of adequate fertiliser, particularly the application of adequate amounts of K and N, is necessary to prevent early senescence, which leads to dead leaves upon which the fungus sporulates (Jellis et al. 1998). The use of adequate amounts of fertilizer, particularly K, has also been shown to reduce susceptibility to infection in field pea (Biddle 2001). K is absorbed by plants in its cation form and it is the major cation in the cytoplasm. It is readily translocated in the phloem and xylem (Marschner 1986). A large number of enzymes are either completely dependent upon or stimulated by K (Suelter 1970). Interestingly, we found that fertigation with low levels of K is associated with higher levels of grey mould in sweet basil; whereas fertigation with comparably higher levels of K can suppress the disease under field conditions (Israeli et al. 2011; Yermiyahu et al. 2013). In grapevine, increased applications of K have been associated with less severe outbreaks of grey mould (Kiraly 1964) and, in pumpkin, increased applications of K have been associated with less severe white mould infections (Abia and Smith 1980).

8.6 Conclusions

As examples drawn from some *Botrytis*-based pathosystems could demonstrate (mentioned above), integrated disease management have advantages over using single means of control and longer durability. A strategy was developed for the integrated biological and chemical control of *B. cinerea* in vegetable crops grown in unheated greenhouses. The integrated strategy, named BOTMAN (Botrytis Manager) has been used in greenhouse-grown tomato and cucumber and achieved a drastic reduction in chemical fungicide use (Shtienberg and Elad 1997) while promoting cultural management and biocontrol in greenhouse-grown vegetable crops. This system takes into consideration not only chemical and biocontrol agents, but also environmental conditions and their potential effects on grey mould epidemics (Shtienberg and Elad 1997). Decisions of whether to spray a biological agent or a fungicide are made in relation to weather forecasts. When cultural means of control seem to be sufficient and slow or no disease progress is expected, the system does not suggest any spray treatments. In contrast, when an outbreak is expected, the application of a chemical fungicide is recommended. In all other cases when the risk is of mid-level, the application of a biocontrol agent is recommended. In this system, disease warnings are based on weather forecasts.

Infections caused by pathogenic *Botrytis* species are more common under high-humidity conditions and cultural techniques aimed at decreasing humidity can suppress those infections. In some cases, cultural methods actually increase the resistance of the crop to the disease. Different cultural methods can be combined for improved disease suppression. Moreover, cultural methods can be combined with chemical botryticides or biocontrol treatments (Chap. 9). Integrated crop management, including disease control, may involve the use a decision-support system, such as BOTMAN.

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